Chemistry Notes By Vasumitra Gajbhiye

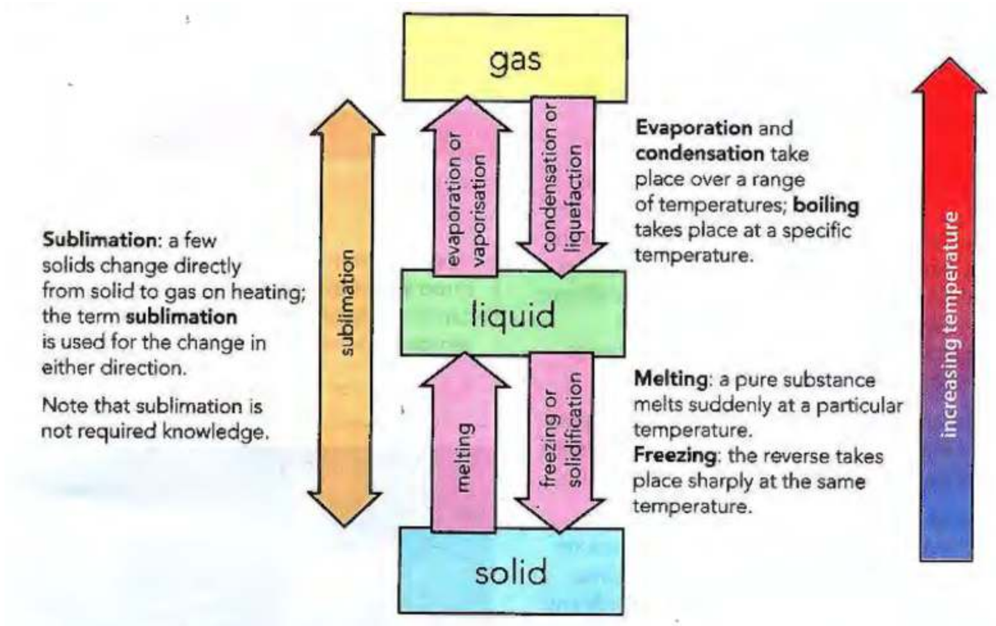
B1. States of Matter

L.1. States of matter

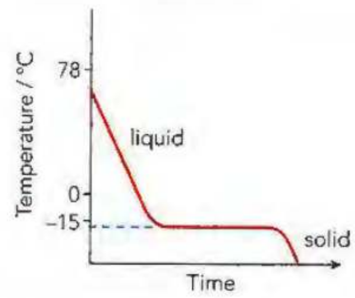
- Differences in the properties of the three states of matter.

Physical state	Volume	Density	Shape	Fluidity
solid	has a fixed volume	high	has a definite shape	does not flow
liquid	has a fixed volume	moderate to high	no definite shape – takes the shape of the container	generally flows easily
gas	no fixed volume – expands to fill the container	low	no definite shape – takes the shape of the container	flows easily

- Changes of physical states and the effect of increasing temperature at atmospheric pressure.



- **Evaporation** → a process occurring at the surface of a liquid, involving the change of state from a liquid into a vapour at a temperature below the boiling point.
- Water is quite a volatile liquid. Ethanol, with a boiling point of 78 °C, is more volatile than water. It has a higher volatility than water and evaporates more easily.
- It is possible, at normal temperatures, to condense a gas into a liquid by increasing the pressure without cooling.
- If the surrounding pressure falls, the boiling point falls. If the surrounding pressure is increased, the boiling point rises.
- A pure substances consists of only one substance without any contaminating impurities. A pure substance melts and boils at definite temperatures.
- This means that we can use BP to test the purity of a sample. These values can also be used to check the identity of an unknown substance.
- Impure substances boil over a range of temperatures. BP increase & MP decrease.
- A cooling Curve show that heat energy is needed to change a solid into a liquid, or a liquid into a gas. When state change from gas to liquid or liquid to solid, heat energy is given out.



- **Particles in Gas are :**
  - arranged totally irregularly
  - spread very far apart
  - able to move randomly
- **Particles in Liquids are :**
  - closely packed together
  - in an irregular arrangement
  - able to move around past each other
- **Particles in Solids are :**
  - packed close together
  - in a regular arrangement or lattice.
  - not able to move freely, but simply vibrate in their fixed positions.
- In a gas, the intermolecular space is large and can be reduced by increasing the external pressure. Therefore, gases are easily compressible. In liquids, this space is very much smaller. As a result, liquids are not very compressible.
- Change in **pressure** of Gas:
  - pressure increases → volume decreases → gas compressed
  - pressure decreases → volume increases → gas expand
- Change in **temperature** of Gas:
  - temperature increase → volume increase → gas expand
  - temperature decrease → volume decrease → gas compressed
- An increase in temperature increases the kinetic energy of each particle, as the thermal energy is transformed to kinetic energy, so they move faster and there is less chance of interaction between them. The gas particles move more freely and occupy a greater volume.

- An decrease in temperature decrease the kinetic energy of each particle, as the kinetic energy is transformed to thermal energy, the particles are moving more slowly. They are more likely to interact with each other and move together to occupy a smaller volume.
- An increase in pressure pushes the particles closer together meaning that the moving particles are more likely to interact with each other and move closer together.
- An decrease in pressure pushes the particles further apart meaning that the moving particles are less likely to interact with each other and further apart together.
- When solids are heated, the particles absorb TE which is converted into KE. This causes its particles to vibrate more and as the temperature increases. The KE is used to break bonds. They vibrate so much that the solid expands until all the bonds are broken and the solid melts.
- On further heating, liquid substance expands more and some particles at the surface gain sufficient energy to overcome the intermolecular forces and evaporate. When the BP is reached, all the particles gain enough energy to escape and the liquids boils

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- **Suspension** → a mixture containing small particles of an insoluble solid, or droplets of an insoluble liquid, spread (suspended) throughout a liquid.
- A concentrated solution contains a high proportion of solute. A dilute solution contains a small proportion of solute.
- Unlike most solids, gases become less soluble in water as the temperature rises.
- Particles move from a region of higher concentration towards a region of lower concentration; eventually, the particles are evenly spread. Their concentration is the same throughout.
- The rate of diffusion in liquids is much slower than in gases.
- Diffusion does not take place in solids as the particles cannot move from place to place.
- The pressure of a gas is the result of collisions of the fast-moving particles with the walls of the container.
- Heavier gas particles move more slowly than lighter particles at the same temperature.
- The average speed of the particles increases with an increase in temperature.

B2. Atoms, Elements and Compounds

L.2. Atomic Structure

- **Element** → a substance that cannot be further divided into simpler substances by chemical methods.
- **Compound** → a substance formed by the chemical combination of two or more elements in fixed proportions.
- **Mixture** → two or more substances mixed together but not chemically combined - the substances can be separated by physical means.

- Structure of an atom → a central nucleus containing neutrons and protons surrounded by electrons in shells.
- Proton number/ atomic number(Z) is the number of protons in the nucleus of an atom.

Subatomic particle	Relative mass	Relative charge	Location in atom
proton	1	+1	in nucleus
neutron	1	0	in nucleus
electron	$\frac{1}{1840}$ (negligible)	−1	outside nucleus

- Mass number/nucleon number(A) is the total number of protons and neutrons in the nucleus of an atom.
- The electrons are held within the atom by an electrostatic force of attraction between them and the positive charge of the protons in the nucleus.
- Hydrogen have only one proton and one electron. It does not have a neutron in its nucleus.
- Group VIII noble gases have a full outer shell. The number of outer shell electrons is equal to the group number in Groups I to VII. The number of occupied electron shells is equal to the period number.

- **Isotopes** are different atoms of the same element that have the same number of protons but different numbers of neutrons.
- Isotopes of the same element have the same chemical properties because they have the same number of electrons and therefore the same electronic configuration.

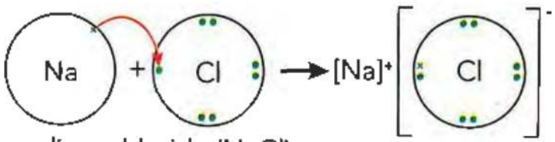
- **Relative atomic mass** → the average mass of naturally occurring atoms of an element on a scale where the carbon-12 atom has a mass of exactly 12 units.
- Some isotopes are radioactive because their nuclei are unstable (radioisotopes).
- <u>The masses of the isotopes of the same element differ and therefore other properties, such as density and rate of diffusion, also vary.</u>

- To calculate relative atomic mass of an element :
    - Calculate the mass of 100 atoms by multiplying the mass of each isotope by its relative abundance.
    - Calculate the average mass of one atom by dividing by 100.
    - Round to required level of accuracy.
- E.g. Lithium have two isotopes, with relative abundance 7.5% and 92.5%. Calculate its relative atomic mass.
    - (6 x 7.5) + (7 x 92.5) = 692.5
    - 692.5 / 100 = 6.925
    - Relative atomic mass of lithium = 6.9 (2 S.F.)

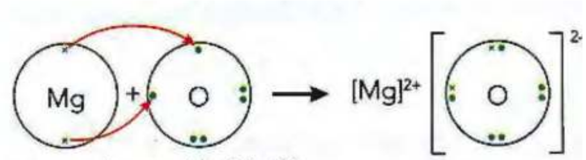
- Group number tell the number of outer shell electrons.
- Period(row) number tell the number of electron shells.

L.3. Chemical Bonding

- Positive ions, known as cations, are formed by loss of electrons.
- Negative ions, known as anions, are formed by gain of electrons.
- An ionic bond is a strong electrostatic attraction between oppositely charged ions.
- When different elements react together to form a compound their characteristic properties are lost, being replaced by those of that compound.
- Solution is a type of mixture.
- To form sodium chloride, sodium atoms loses a electron and chlorine atom gains an electron. Sodium ion has a stable electron arrangement of neon atom. Chloride ion has a stable electron arrangement of argon atom.



sodium chloride (NaCl)



magnesium oxide (MgO)

- Metals are electron donors. Non-metals are electron acceptors.
- Properties of Ionic compounds :

Properties of typical ionic compounds	Reasons for these properties
They have high melting points and boiling points.	Ions are attracted to each other by strong electrostatic forces. Large amounts of energy are needed to separate them.
They are crystalline solids at room temperature.	There is a regular arrangement of the ions in a lattice. Ions with opposite charge are next to each other.
They are often soluble in water (not usually soluble in organic solvents, e.g. ethanol, methylbenzene).	Water is attracted to charged ions and therefore many ionic solids dissolve.
They conduct electricity when molten or dissolved in water (not when solid).	In the liquid or solution, the ions are free to move about. They can move towards the electrodes when a voltage is applied.

- Ionic compounds do not conduct electricity when solid because there are no free mobile electrons or ions to carry the current through the solid.
- Giant lattice structure of a ionic compound is a regular arrangement of alternating positive and negative ions

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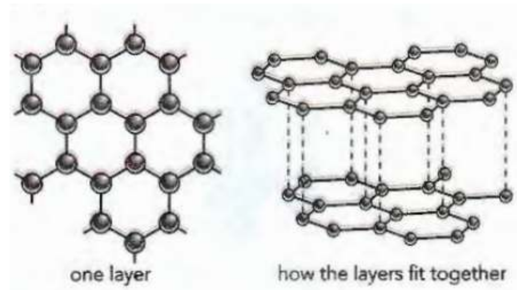
- A covalent bond is formed when a pair of electrons is shared between two atoms leading to noble gas electronic configurations.
- Diatomic molecules contain two atoms, doesn't matter that they should be of the same element. E.g. H_2 and CO.
- Properties of Covalent compounds + explanation:
    - **Low melting points and boiling points** → They are made of simple covalent molecules. The forces between the molecules (intermolecular forces) are only very week. Not much energy is needed to move the molecules further apart.
    - **Poor electrical conductivity** → There are no free electrons or ions present to carry the electrical current.

- In Diamond each carbon atom is attached to four other atoms - the atoms are arranged tetrahedrally.

- Diamond doesn't conduct electricity because all the electrons are used to form covalent bonds.
- Diamond is used in cutting tools because diamond is very hard. Diamond is very hard because strong covalent bonds extends throughout the whole structure.
- In graphite, carbon atoms are arranged in flat layers of linked hexagons.
- Each graphite layer is a 2-D giant molecule.

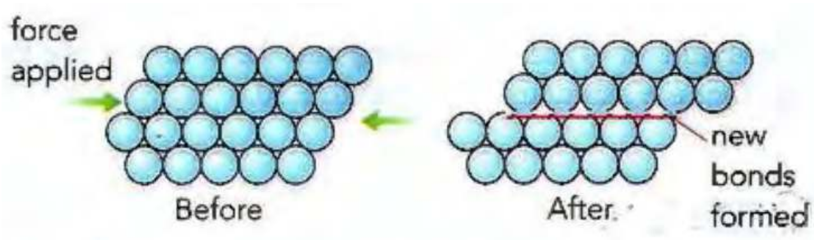


- Each carbon atom is bonded to three others by strong covalent bonds. Between the layers there are weaker forces of attraction.
- This means that graphite feels slippery and can be used as a lubricant. Graphite is used in pencils. When we write with a pencil, thin layers of graphite are left stuck to the paper.
- Graphite can conduct electricity because it has free electrons.
- In silica (silicon(IV) dioxide, SiO_2) each silicon atom is bonded to four oxygen atoms, but each oxygen is only bonded to two silicon atoms.
- Similar properties of diamond & silicon(IV) oxide:
    - tetrahedral arrangement of atoms.
    - strong covalent bonds throughout
    - very hard
    - high MP and BP
    - does not conduct electricity, because there are no free electrons to move through the structure.

- Metallic bonding is the electrostatic attraction between the positive ions in a giant metallic lattice and a 'sea' of delocalised electrons.
- Properties of giant metallic lattices + explanation:
    - **High MP & BP** → Large amount of energy is needed to overcome the strong and extensive force of attraction between the positive metal ions and the 'sea' of

delocalised electrons moving within the lattice.

  - **Good electrical conductivity** → In metal electrons are free/ delocalised, so mobile electrons can move through the structure, carrying the current.
  - **Malleability and ductility** → Metals are easily bent and shaped (malleable) or stretched into wires (ductile). The positive ions in a metal are arranged in layers. When a force is applied, the layers can slide over each other. The attractive forces in metallic bonding act in all directions to hold the structure together. This means that when the layers slide over each other new bonds are easily formed. This movement of layers leaves the metal with a different shape.



B3. Stoichiometry

L.4. Chemical Formulae and Equations

- **Molecular formula** → the number and type of different atoms in one molecule.
- **Empirical formula** → the simplest whole number ratio of the different atoms or ions in a compound.
- **Rules while naming compounds:**
  - If there is a metal in the compound, it is named first
  - Where the metal can form more than one ion, then the name indicates which ion is present. Ex - iron(II) chloride.
  - Compounds containing only two elements have names ending in -ide, except hydroxides. Ex - sodium hydroxide.
  - Compounds containing a compound ion (usually containing oxygen) have names that end with -ate. Ex - calcium carbonate.

- Law of conservation of mass states that the total mass of all the products of a chemical reaction is always equal to the total mass of all the reactants.
- alkali metal + water → <u> metal hydroxide</u> + hydrogen.
    - Ex - potassium + water → potassium hydroxide + hydrogen
- To write a ionic equation:
    1. Write down the equation: $HCl(aq) + NaOH(aq) => NaCl(aq) + H_2O(l)$
    2. Writing out all the ions present, we get:

$$[H^+(aq) + \cancel{Cl^-(aq)}] + [\cancel{Na^+(aq)} + OH^-(aq)] \rightarrow [\cancel{Na^+(aq)} + \cancel{Cl^-(aq)}] + H_2O(l)$$

    3. Write the remaining equation: $H^+(aq) + OH^-(aq) => H_2O(l)$

- Relative atomic mass, A_r, is the average mass of the isotopes of an element compared to 1/12th of the mass of an atom of C-12.
- Relative molecular mass, M_r, is the sum of the relative atomic masses. Relative formula mass, M_r, will be used for ionic compounds

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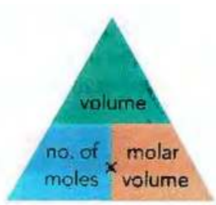
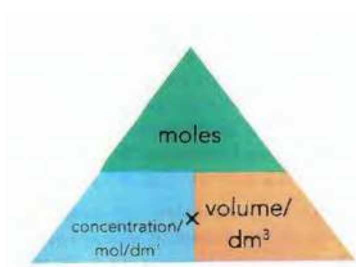
- The mole, mol, is the unit of amount of substance and one mole contains 6.02 × 10^23 particles, e.g. atoms, ions, molecules; this number is the Avogadro constant

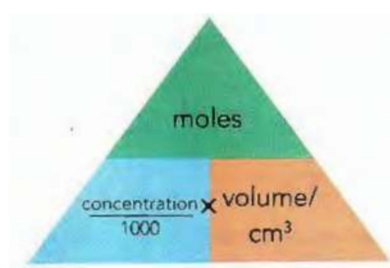
$$\text{amount of substance (mol)} = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

- The molar gas volume, is taken as $24dm^3$ at room temperature and pressure, r.t.p.

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{predicted yield}} \times 100 \qquad \text{percentage purity} = \frac{\text{mass of pure product}}{\text{mass of impure product}} \times 100$$



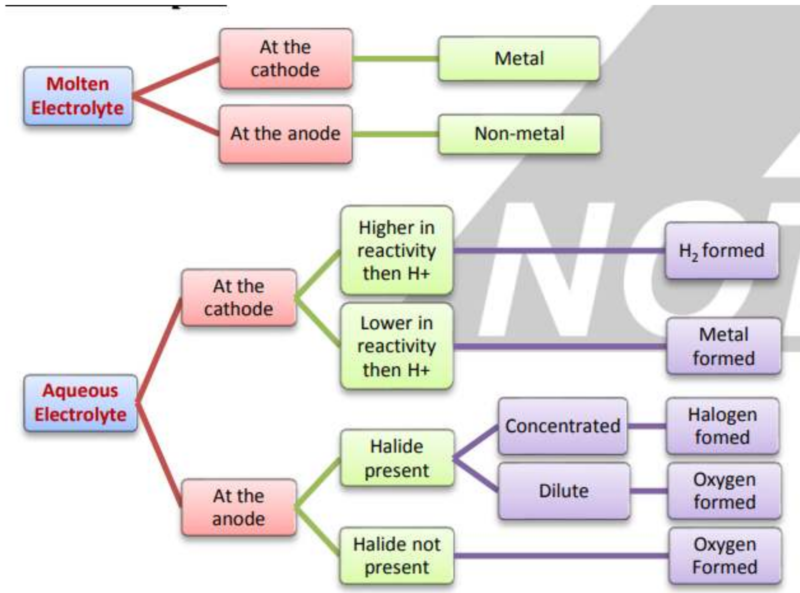




B4. Electrochemistry

L.6. Electrochemistry

- Electrolysis is the decomposition of an ionic compound, when molten or in aqueous solution, by the passage of an electric current.
- The anode is the positive electrode
- The cathode is the negative electrode
- The electrolyte is the molten or aqueous substance that undergoes electrolysis
- Metals or hydrogen are formed at the cathode and non-metals (other than hydrogen) are formed at the anode.
- Metal objects are electroplated to improve their appearance and resistance to corrosion.
- Object to be electroplated is placed at the cathode. And the metal from which it is electroplated placed at the anode. ions travel from anode to cathode. Nitrate of the metal at the anode is used as electrolyte.
- The electrons move in the external circuit
- The loss of electrons occur at anode and gain of electrons occur at the cathode.
- Oxidation occurs at the anode and reduction occurs at the cathode.
- The ions move in the electrolyte.
- Electrolysis of aqueous copper(ll) sulfate using carbon electrode produce copper at the cathode and oxygen at anode.
- Electrolysis of aqueous copper(ll) sulfate using copper electrode produce copper at the cathode and anode dissolve to give copper ions in the electrolyte. Mass of cathode increase and mass of anode decrease.
- Electrolysis of sulfuric acid produce oxygen at the anode and hydrogen at the cathode.



- A hydrogen–oxygen fuel cell uses hydrogen and oxygen to produce electricity with water as the only chemical product.

Advantages	Disadvantages
• renewable if produced using solar or wind energy • lower flammability than petrol • virtually emission-free • zero emissions of CO_2 • non-toxic	• non-renewable if generated using nuclear energy or energy from fossil fuels • large fuel tank required • as yet there are very few 'filling stations', where a car could be topped up with hydrogen • engine redesign needed or a fuel cell system • currently expensive

B5. Chemical energetics

L.7. Chemical energetics

- An exothermic reaction transfers thermal energy to the surroundings leading to an increase in the temperature of the surroundings. ΔH is negative for exothermic reactions

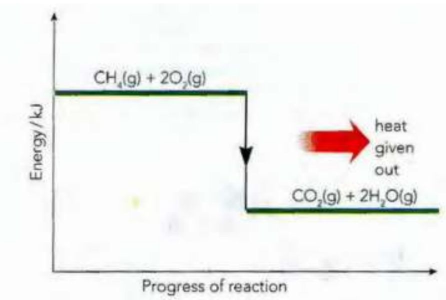


Figure 7.9: Reaction pathway diagram for the exothermic reaction between methane and oxygen.

- An endothermic reaction takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings. ΔH is positive for endothermic reactions

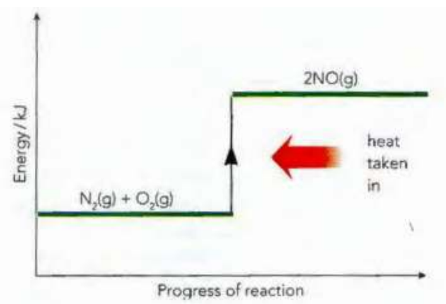


Figure 7.10: A reaction pathway diagram for the reaction between nitrogen and oxygen.

- The transfer of thermal energy during a reaction is called the enthalpy change, ΔH, of the reaction.
- Activation energy, E_a, is the minimum energy that colliding particles must have to react.
- Bond breaking is an endothermic process and bond making is an exothermic process.

B6. Chemical reactions

L.8. Rates Of Reactions

Physical Change	Chemical Change
When a substance undergoes a physical change, its composition remains the same despite its molecules being rearranged.	When a substance undergoes a chemical change, its molecular composition is changed entirely. Thus, chemical changes involve the formation of new substances.
Physical change is a temporary change.	A chemical change is a permanent change.
A Physical change affects only physical properties i.e. shape, size, etc.	Chemical change both physical and chemical properties of the substance including its composition
A physical change involves very little to no absorption of energy.	During a chemical reaction, absorption and evolution of energy take place.
Some examples of physical change are freezing of water, melting of wax, boiling of water, etc.	A few examples of chemical change are digestion of food, burning of coal, rusting, etc.
Generally, physical changes do not involve the production of energy.	Chemical changes usually involve the production of energy (which can be in the form of heat, light, sound, etc.)
In a physical change, no new substance is formed.	A chemical change is always accompanied by one or more new substance(s).
Physical change is easily reversible i.e original substance can be recovered.	Chemical changes are irreversible i.e. original substance cannot be recovered.

- Increasing the concentration of solutions increase the rate of reaction because there are more number of particles per unit volume. So frequency of collision between particles increase.
- Increasing the pressure of a gas increase the rate of reaction because there are more number of particles per unit volume. So frequency of collision between particles increase.
- Increasing the surface area of a solid increase the rate of reaction because more particles are exposed to collide. So frequency of collision between particles increase.
- Increasing the temperature increase the rate of reaction because there is an increase in average kinetic energy of the particles. particle move faster. more particles have energy greater than activation energy. So frequency of collision between particles increase.

- Adding a catalyst, including enzymes, increase rate of reaction. Decreases the activation energy of the reaction.
- A catalyst increases the rate of a reaction and is unchanged at the end of a reaction.

L.9. Reversible reactions and Equilibrium

- Some chemical reactions are reversible and shown by the symbol $\rightleftharpoons$.
- Anhydrous copper(II) sulfate react with water to form hydrated copper(II) sulfate. There is a colour change from white to blue. The reaction is exothermic.
- Anhydrous cobalt(II) chloride react with water to form hydrated cobalt(II) chloride. There is a colour change from blue to pink. The reaction is exothermic.
- A reversible reaction in a closed system is at equilibrium when:
  - The rate of the forward reaction is equal to the rate of the reverse reaction.
  - The concentrations of reactants and products are no longer changing.
- In equilibrium, increasing the temperature favours the side with endothermic reaction.
- In equilibrium, decreasing the temperature favours the side with exothermic reaction.
- In equilibrium, increasing the pressure favours the side with fewer mols of gas.
- In equilibrium, decreasing the pressure favours the side with more mols of gas.
- In equilibrium, increasing the concentration of a reactant favours the side that makes less of that reactant, i.e. favours the opposite side of the reaction.
- In equilibrium, adding catalysts increases the rate of both reactions.
- Reaction for Haber process → $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
- The sources of the hydrogen is methane/ hydrocarbon and for nitrogen is air in the Haber process.
- The typical conditions in the Haber process are 450°C, 20000kPa /200atm and an iron catalyst.
- Reaction for conversion of sulfur dioxide to sulfur trioxide in contact process → $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$
- The sources of the sulfur dioxide is burning sulfur or roasting sulfide ores and oxygen is air in the Contact process.

- The typical conditions for the conversion of sulfur dioxide to sulfur trioxide in the Contact process is 450°C, 200kPa /2atm and vanadium(V) oxide catalyst.

L.10. Redox Reactions

- Roman numeral are used to indicate the oxidation number of an element in a compound.
- Redox reactions involve simultaneous oxidation and reduction.
- Oxidation is gain of oxygen or increase in oxidation state or loss of electrons
- Reduction is loss of oxygen or decrease of oxidation state or gain of electrons.
- Redox reactions involves gain and loss of oxygen.
- The oxidation number of elements in their uncombined state is zero
- The oxidation number of a monatomic ion is the same as the charge on the ion.
- The sum of the oxidation numbers in a compound is zero.
- The sum of the oxidation numbers in an ion is equal to the charge on the ion.
- Potassium iodide is a reducing agent and is used to test for oxidising agent. There is a colour change from colourless to yellow-brown because iodide ions are oxidised to iodine. If starch indicator is added there is a colour change to blue-black because of the presence of iodine.
- Acidified potassium manganate(VII) is an oxidising agent and used to test for reducing agent. There is a color change from purple to colourless because manganate(VII) ions are reduced.
- Fe(III) → Fe(II) is reduction. and Fe(II) → to Fe(III) is oxidation.
- An oxidising agent is a substance that oxidises another substance and is itself reduced.
- A reducing agent is a substance that reduces another substance and is itself oxidised.

B7. Acids, Bases and Salts

L.11. Acids and Bases

- Metal + acid ⇒ salt + hydrogen

- Acid + base ⇒ salt + water
- Metal carbonate + acid ⇒ salt + water + carbon dioxide

Indicator	Colour in acid	Neutral colour	Colour in alkali
litmus	red	purple	blue
thymolphthalein	colourless	colourless	blue
methyl orange	red	orange	yellow

- Bases are oxides or hydroxides of metals and alkalis are soluble bases.
- Base + ammonium salt ⇒ salt + water + ammonia
- Aqueous solutions of acids contain H^+ ions and aqueous solutions of alkalis contain OH^- ion.
- Acids are proton donors and bases are proton acceptors
- Strong acid → An acid that is completely <u>dissociated</u> in aqueous solution.
- Weak acid → An acid that is <u>partially dissociated</u> in aqueous solution.
- Hydrochloric acid is a strong acid, as shown by the symbol equation, $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$
- <u>Ethanoic acid is a weak acid, as shown by the symbol equation,</u> $\underline{CH3COOH(aq) \rightleftharpoons H^+(aq) + CH3COO^-(aq)}$
- Metal oxides are basic oxides. Non-metal oxides are acidic oxides.
- Amphoteric oxides → oxides that react with acids and with bases to produce a salt and water. E.g. Al_2O_3 and ZnO

L.12. Preparation of salts

- **Making salt by acid + excess metal**
    1. Warm the acid. Switch off the Bunsen burner. Add an excess of the metal to the acid. Wait until no more bubbles.
    2. Filter the solution
    3. Heat the filtrate till crystallisation point. leave it to cool. Filter the crystals, wash, dry between filter paper.

- **Making salt by acid + excess insoluble base (metal oxide, metal hydroxide)**
    1. Add excess insoluble base to an acid. stir. Wait until the solution no longer turns red blue litmus paper blue.
    2. Filter the solution
    3. Heat the filtrate till crystallisation point. leave it to cool. Filter the crystals, wash, dry between filter paper.
- **Making salt by acid + excess metal carbonate**
    1. Add an excess of the metal carbonate to the acid. Wait until no more bubbles.
    2. Filter the solution
    3. Heat the filtrate till crystallisation point. leave it to cool. Filter the crystals, wash, dry between filter paper.
- **<u>Titration</u>**
    1. measure known volume of alkali into a conical flask
    2. using pipette
    3. methyl orange indicator added
    4. add hydrochloric acid
    5. from a burette in small portions, and stir as each portion is added
    6. until indicator changes colour
    7. record / calculate volume acid added
    8. repeat with same volume acid without indicator
- **General solubility rules for salts:**
    - sodium, potassium and ammonium salts are soluble
    - nitrates are soluble
    - chlorides are soluble, except lead and silver
    - sulfates are soluble, except barium, calcium and lead
    - carbonates are insoluble, except sodium, potassium and ammonium
    - hydroxides are insoluble, except sodium, potassium, ammonium and calcium (partially)

- A hydrated substance is a substance that is chemically combined with water.
- An anhydrous substance is a substance containing no water.
- **Precipitations reaction**
    1. add equal volume of water to equal mass of both the soluble salts in two different beakers
    2. mix the two aqueous salt solutions.
    3. filter the precipitate
    4. wash under running water, dry between filter paper
- Water of crystallisation as the water molecules present in hydrated crystals, including $CuSO_4 \cdot 5H_2O$ and $CoCl_2 \cdot 6H_2O$

B8. The Periodic Table

L.13. The Periodic Table

- The Periodic Table is an arrangement of elements in periods and groups and in order of increasing proton number/ atomic number.
- Elements become less metallic as you move to the right of the periodic table.
- Group number (column) represent the valency of the element.
- Period number (row) represent the number of electron shells in an element.
- Elements in same group of the periodic table have similar chemical properties because they have <u>same number of valence electrons, and similar electronic configuration.</u>
- As you move down group 1:
    - decreasing melting point and boiling point
    - increasing density
    - increasing reactivity
- Group I alkali metals, lithium, sodium and potassium, as relatively soft metals.
- As you move down group 7:

    - increasing density
    - decreasing reactivity
    - (get darker in colour)
- Colour of halogens at r.t.p. :
    - fluorine, a pale yellow gas
    - chlorine, a pale yellow-green gas
    - bromine, a red-brown liquid
    - iodine, a grey-black solid
- More reactive halogens displace less reactive halogens from their compound.
- Transition metals:
    a. have high densities
    b. have high melting points
    c. form coloured compounds
    d. often act as catalysts as elements and in compounds
    e. have variable oxidation states.
- Group VIII noble gases are unreactive, monatomic gases because they have full outer shell.

B9. Metals

L.14. Metallic elements and Alloys

Physical property	Metals	Non-metals
melting points and boiling points	Metals are usually solids (except for mercury, which is a liquid) at room temperature. Many metals have high melting and boiling points.	Non-metals are solids or gases at room temperature (except for bromine, which is a liquid). Their melting and boiling points are often low.
electrical conductivity	All metals are good conductors of electricity.	Non-metals are poor conductors of electricity (except for graphite). They tend to be insulators.
thermal conductivity	Metals are good conductors of heat.	Non-metals are generally poor thermal conductors (except diamond, which conducts heat strongly).
malleability and ductility	The shape of a piece of metal can be changed by hammering (they are malleable). They can also be pulled out into wires (they are ductile).	Non-metals are not malleable; they are brittle and break easily when hit.
strength and hardness	Metals are usually strong and dense. When a force is applied, they are hard and do not shatter (they are not brittle).	Most non-metals are softer than metals (but diamond is very hard). Their densities are often low.
ability to produce a sound	Metals usually make a ringing sound when struck (they are **sonorous**).	Non-metals are not sonorous.
colour and appearance	Metals are grey in colour (except gold and copper). They can be polished.	Non-metals vary in colour. They often have a dull surface when solid.

Metal	Reaction with ...		
	Air	Water	Dilute HCl
sodium	burn very strongly in air to form oxide	react with cold water to give hydrogen	react very strongly to give hydrogen
calcium			
magnesium			
aluminium[a]	burn less strongly in air to form oxide	react with steam, when heated, to give hydrogen	react less strongly to give hydrogen
zinc			
iron			
lead	react slowly to form oxide layer when heated	do not react	do not react
copper			
silver	do not react	do not react	do not react
gold			

- Metals react with oxygen to form metal oxides. Less reactive metals like silver and gold don’t react with oxygen.
- Metal + water:
    - More reactive metals like sodium, potassium and calcium react with cold water.

  - Magnesium reacts only slowly with cold water.
  - a much more vigorous reaction takes place if steam is passed over heated magnesium, iron or zinc.
  - if a metal reacts with cold water, a metal hydroxide and hydrogen are formed
  - if a metal reacts only with steam, then a metal oxide is formed.
- Moderately reactive metals such as magnesium, zinc or iron can be reacted safely with dilute acids to produce hydrogen gas.
- Metal less reactive than hydrogen, don’t react with acids.
- **Uses of metals:**
  - aluminium in the manufacture of aircraft because of its low density
  - aluminium in the manufacture of overhead electrical cables because of its low density and good electrical conductivity
  - aluminium in food containers because of its resistance to corrosion
  - copper in electrical wiring because of its good electrical conductivity and ductility and it is unreactive.
- An alloy is a mixture of a metal with other elements.
- Brass is a mixture of copper and zinc.
- Stainless steel is a mixture of iron and other elements such as chromium, nickel and carbon.
- Alloys can be harder and stronger than the pure metals and are more useful.
- Alloys are harder than pure metals because the different sized atoms in alloys mean the layers can no longer slide over each other.
- Stainless steel is used in cutlery because of its hardness and resistance to rusting.

L.15. Reactivity of Metals

- **Reactivity series** → potassium, sodium, calcium, magnesium, aluminium, carbon, zinc, iron, hydrogen, copper, silver, gold
- **Experiment to deduce order of reactivity of metals :**
  - add metals to HCl in a beaker / flask / test-tube

- **fair test – max 4**
  • same volume HCl
  • same concentration HCl
  • same temperature acid
  • same mass / moles / amount metals
  • same surface area / form of metal
- **measure**
  • start timing when solid added
  • stop timing when all solid gone / reaction to stop
- **OR**
  • start timing when solid added
  • stop timing when collected set volume of gas
- **OR**
  • start timing when solid added
  • measure volume of gas after a set time
- **OR**
  • measure temperature at start
  • measure temperature after reaction OR highest temperature reached
- **conclusion**
  • shortest time is most reactive
- **OR**
  • biggest volume of gas most reactive
- **OR**
  • biggest temperature change most reactive

* More reactive metals displace less reactive metals because <u>more reactive metals have greater tendency to form positive ions.</u>
* Aluminium has a thin layer of aluminium oxide. This layer sticks to the surface of the metal and doesn't flakes off. This layer makes aluminium unreactive.

L.16. Extraction and corrosion of Metals

* Metals below carbon in the series can be extracted by heating their oxides with carbon, but those above carbon must be extracted by electrolysis.

- **Extraction of Iron from hematite:**
  - The burning of carbon (coke) to provide heat and produce carbon dioxide
    - $C + O_2 => CO_2$
  - The reduction of carbon dioxide to carbon monoxide
    - $CO_2 + C => 2CO$
  - The reduction of iron(III) oxide by carbon monoxide
    - $Fe_2O_3 + 3CO => 2Fe + 3CO_2$
  - The thermal decomposition of calcium carbonate /limestone to produce calcium oxide
    - $CaCO3 => CaO + CO_2$
  - The formation of slag
    - $CaO + SiO_2 => CaSiO_3$
- Main ore for aluminium is bauxite and it is extracted by electrolysis.
- **Extraction of aluminium:**
  - Bauxite is purified to aluminium oxide
  - Cryolite is added
    - Cryolite reduces the melting point of aluminium oxide and increases conductivity of aluminium oxide
  - Electrolysis of this mixture is done
    - At Cathode → $Al^3(l) + 3e^- => Al(l)$
    - At Anode → $2O^2 - (l) => O_2 + 4e^-$
- At the high temperature in the cell, the oxygen reacts with the carbon in the electrodes forming carbon dioxide. The carbon anodes slowly burn away and therefore have to be replaced frequently.

- Water and oxygen is required for the rusting of iron.
- Painting, greasing and coating with plastic are some common barrier methods.
- Barrier methods prevent rusting by preventing iron and steel from coming into contact with water and oxygen.

- The use of zinc in galvanising is an example of a barrier method and sacrificial protection.
- Zinc is electroplated on iron for galvanisation.
- Zinc is more reactive than iron so loses electrons more easily. Hence, it corrodes in preference to iron.

B10. Chemistry of the environment

L.17. Chemistry of the environment

- Test for water:
    - Anhydrous copper(II) sulfate turn from white to blue
    - Anhydrous cobalt(II) chloride turns from blue to pink
- To check the purity of water check it's boiling or the ice's melting point.
- Distilled water is used in practical chemistry rather than tap water because it contains fewer chemical impurities.
- Water from natural sources may contain substances, including:
    1. dissolved oxygen
    2. metal compounds
    3. plastics
    4. sewage
    5. harmful microbes
    6. nitrates from fertilisers
    7. phosphates from fertilisers and detergents
- Some of these substances are beneficial, including:
    1. dissolved oxygen for aquatic life
    2. some metal compounds provide essential minerals for life
- Some of these substances are potentially harmful, including:

1. some metal compounds are toxic
2. some plastics harm aquatic life
3. sewage contains harmful microbes which cause disease
4. nitrates and phosphates lead to deoxygenation of water and damage to aquatic life

- Purification of water:

1. sedimentation and filtration to remove solids
2. use of carbon to remove tastes and odours
3. chlorination to kill microbes
4. Distillation is not performed because chlorination does the job of distillation and also have few more beneficial effects.

- Ammonium salts and nitrates are used in fertilisers.
- NPK fertilisers are used to provide the elements nitrogen, phosphorus and potassium for improved plant growth.

- The composition of clean, dry air is approximately 78% nitrogen, 21% oxygen and the remainder as a mixture of noble gases and carbon dioxide.
- Source of some air pollutants:

1. carbon dioxide from the complete combustion of carbon-containing fuels
2. carbon monoxide and particulates from the incomplete combustion of carbon-containing fuels
3. methane from the decomposition of vegetation and waste gases from digestion in animals
4. oxides of nitrogen from car engines
5. sulfur dioxide from the combustion of fossil fuels which contain sulfur compounds

- Adverse effect of these air pollutants:

1. carbon dioxide: higher levels of carbon dioxide leading to increased global warming, which leads to climate change
2. carbon monoxide: toxic gas
3. particulates: increased risk of respiratory problems and cancer

4. methane: higher levels of methane leading to increased global warming, which leads to climate change
5. oxides of nitrogen: acid rain, photochemical smog and respiratory problems
6. sulfur dioxide: acid rain

- The sun emits infrared radiation towards the earth. Earth reflect the infrared radiation. Greenhouse gases absorb this infrared radiation. This reduce the thermal energy loss to the space. If excess thermal energy is absorbed this causes enhanced green house effect and increases the average temperature of earth and cause climate change.
- Strategies to reduce the effects of these environmental issues:

1. **climate change:** planting trees, reduction in livestock farming, decreasing use of fossil fuels, increasing use of hydrogen and renewable energy, e.g. wind, solar
2. **acid rain:** use of catalytic converters in vehicles, reducing emissions of sulfur dioxide by using low-sulfur fuels and flue gas desulfurization with calcium oxide

- Photosynthesis is the reaction between carbon dioxide and water to produce glucose and oxygen in the presence of chlorophyll and using energy from light.
- carbon dioxide + water → glucose + oxygen
- $6CO_2 + 6H_2O => C_6H12O_6 + 6O_2$
- Oxides of nitrogen are removed from car using catalytic converter.
- $2CO + 2NO => 2CO_2 + N_2$

B11. Organic chemistry

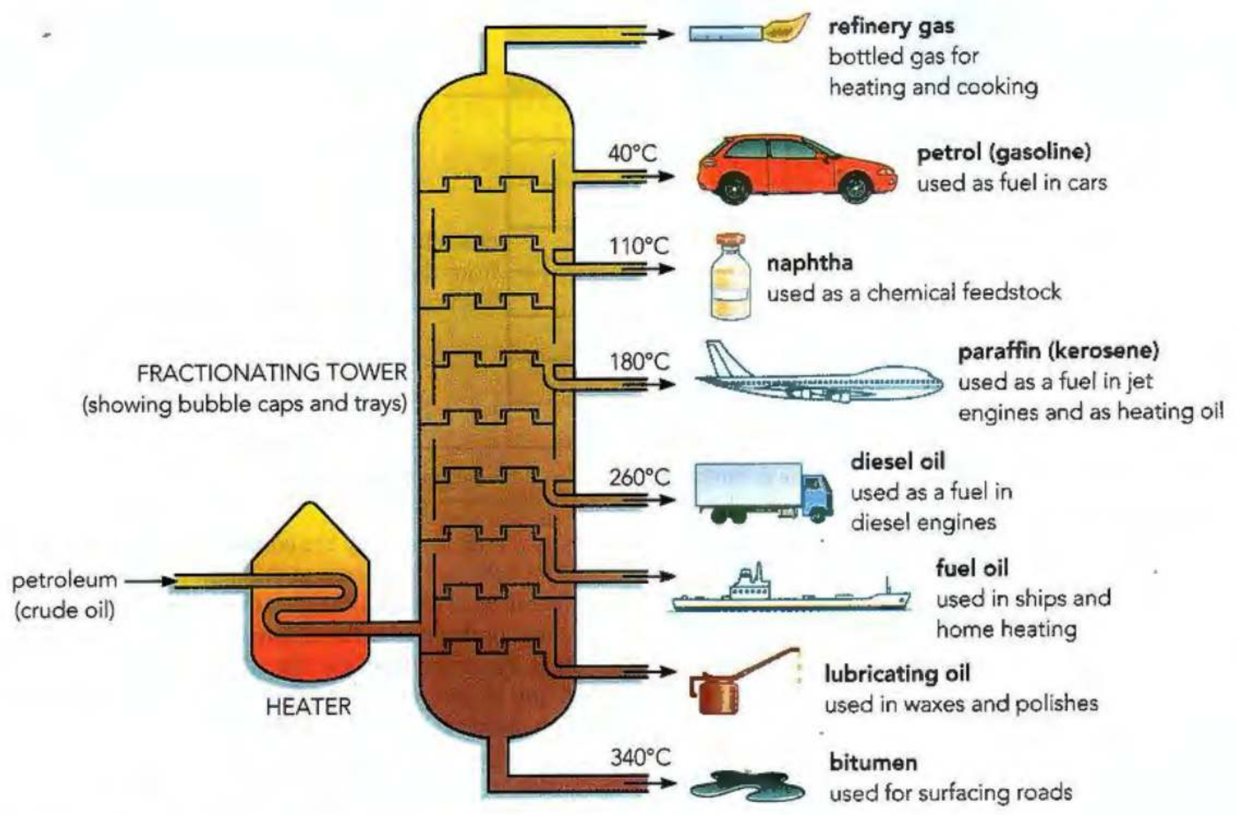
L.18. TO L.20.

- General Formulas:
  - Alkane → CnH2n+2
  - Alkene → CnH2n
  - Alcohol → CnH2n+1OH
  - Carboxylic acid → CnH2n+1COOH

- **Functional group** → an atom or group of atoms that determine the chemical properties of a homologous series.
- A homologous series is a family of similar compounds with similar chemical properties due to the presence of the same functional group.
- A saturated compound has molecules in which all carbon–carbon bonds are single bonds.
- An unsaturated compound has molecules in which one or more carbon–carbon bonds are not single bonds.
- Structural formula is an unambiguous description of the way the atoms in a molecule are arranged, e.g. $CH_2 = CH_2, CH_3CH_2OH, CH_3COOCH_3$
- Structural isomers is compounds with the same molecular formula, but different structural formulae,

  e.g. C_4H10 as $CH_3CH_2CH_2CH_3$ and
  $CH_3CH(CH_3)CH_3$ and C_4H_8 as $CH_3CH_2CH = CH_2$
  and $CH_3CH = CHCH_3$
- The general characteristics of a homologous series are:
  1. having the same functional group
  2. having the same general formula
  3. differing from one member to the next by a –CH_2– unit
  4. displaying a trend in physical properties
  5. sharing similar chemical properties

- Fossil fuels are coal, natural gas and petroleum.
- Methane is the main constituent in natural gas.
- Hydrocarbon are compounds that contain hydrogen and carbon only.
- Petroleum is a mixture of hydrocarbons.



- As you move from the bottom to the top of the fractionating column:
  - decreasing chain length
  - higher volatility
  - lower boiling points
  - lower viscosity
- Uses of fractions:
  - refinery gas fraction for gas used in heating and cooking
  - gasoline /petrol fraction for fuel used in cars
  - naphtha fraction as a chemical feedstock
  - kerosene /paraffin fraction for jet fuel
  - diesel oil/ gas oil fraction for fuel used in diesel engines
  - fuel oil fraction for fuel used in ships and home heating systems
  - lubricating oil fraction for lubricants, waxes and polishes

- bitumen fraction for making roads

- The bonding in alkanes is single covalent and alkanes are saturated hydrocarbons.
- The properties of alkanes are being generally unreactive, except in terms of combustion and substitution by chlorine.
- In a substitution reaction one atom or group of atoms is replaced by another atom or group of atoms.
- The substitution reaction of alkanes with chlorine is a photochemical reaction, with ultraviolet light providing the activation energy. Only one hydrogen atom is substituted. hydrogen chloride is one of the product.

- The bonding in alkenes includes a double carbon–carbon covalent bond and alkenes are unsaturated hydrocarbons.
- Large change alkanes are cracked to small change alkenes and hydrogen in the presence of a catalyst and high temperature (~500°C).
- All cracking reactions give:
  - an alkane with a shorter chain than the original, and a short-chain alkene
  - or two or more alkenes and hydrogen.
- The demand for the various fractions from the refinery does not necessarily match with their supply from the oil fields. There is a greater demand for the lighter fractions such as petrol that have shorter chain length. The longer molecules from these heavier fractions can, be broken into shorter, more valuable, alkane or alkene molecules using cracking.
- To distinguish between saturated and unsaturated hydrocarbons add bromine water to it. If the colour remains yellow-orange it is saturated and if the colour change to colourless it is unsaturated.
- In an addition reaction only one product is formed.
- During combustion of alkene, carbon dioxide and water are produced if there is sufficient supply of oxygen. If not carbon monoxide is produced.
- Alkenes react with bromine in addition reaction. The double bond break open and a bromine molecule attach to the alkene. And it becomes bromo-alkane(NOT bromo-alkene).

- Alkenes react with hydrogen to form alkane. The addition of hydrogen across a carbon-carbon double bond is known as hydrogenation. Ethene reacts with hydrogen if the heated gases are passed together over a nickel catalyst. The unsaturated ethane is the product.
- Conditions for hydrogenation:
  - heated nickel catalyst
  - 150°C to 300°C temperature.
  - heated gas.

- **Manufacture of Ethanol:**
  - fermentation of aqueous glucose at 25–35°C in the presence of yeast and in the absence of oxygen.
  - catalytic addition of steam to ethene at 300°C and 6000kPa /60 atm in the presence of an acid catalyst.
- Combustion of ethanol gives carbon dioxide and water.
- $C_2H_5OH + 3O_2 => 2CO_2 + 3H_2O$
- Ethanol is used as a solvent and as a fuel.

Ethanol by the catalytic hydration of ethene	Ethanol by fermentation
originates from a non-renewable resource (petroleum)	made from readily renewable resources
small-scale equipment capable of withstanding pressure	relatively simple, large vessels
a continuous process	a batch process – need to start process again each time
a fast reaction rate	a relatively slow process
yields highly pure ethanol	ethanol must be purified by subsequent distillation – though fermented product can be used as it is for some purposes
a sophisticated, complex method	a simple, straightforward method

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- Ethanoic acid + metal ⇒ salt + hydrogen

ethanoic acid + magnesium → magnesium ethanoate + hydrogen

$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$

- Ethanoic acid + base ⇒ salt + water

$$\underset{\text{acid}}{\text{ethanoic}} + \underset{\text{hydroxide}}{\text{sodium}} \rightarrow \underset{\text{ethanoate}}{\text{sodinm}} + \text{water}$$

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

- Ethanoic acid + carbonate ⇒ salt + water + carbon dioxide

$$\underset{\text{carbonate}}{\text{calcium}} + \underset{\text{acid}}{\text{ethanoic}} \rightarrow \underset{\text{ethanoate}}{\text{calcium}} + \text{water} + \underset{\text{dioxide}}{\text{carbon}}$$

$$CaCO_3 + 2CH_3COOH \rightarrow (CH_3COO)_2Ca + H_2O + CO_2$$

- **Notice** that in formula ethanoate group comes first then the metal.
- **Formation of ethanoic acid:**
  - It is produced commercially from wine by biochemical oxidation using bacteria (Acetobacter). The production of wine vinegar is an example of traditional biotechnology. The bacteria used are naturally present in the air and wine can simply become 'vinegary' if it is left open to the air.
  - Strong oxidising agent (acidified potassium manganate(VII)) is added to ethanol.

$$\text{ethanol} + \underset{\text{from oxidising agent}}{\text{oxygen}} \rightarrow \text{ethanoic acid} + \text{water}$$

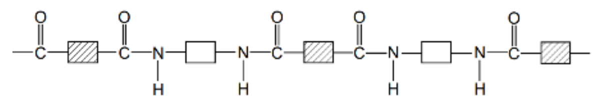
$$C_2H_5OH + 2[O] \rightarrow CH_3COOH + H_2O$$

- Carboxylic acid react with alcohol in the presence of acid catalyst (concentrated sulfuric acid) to from ester. alcohol form the first part of the name (alkyl group) and carboxylic acid form the second part of the name (-notate group).

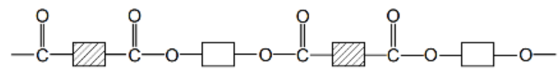
- Polymers is large molecules built up from many smaller molecules called monomers.
- The formation of poly(ethene) is an example of addition polymerisation using ethene monomers.
- Plastics are made from polymers.

	Addition polymerisation	Condensation polymerisation
monomers used	usually many molecules of a single monomer	molecules of two monomers usually used
	monomer is unsaturated, usually contains a C=C double bond	monomers contain reactive functional groups at ends of each molecule
reaction taking place	an addition reaction – monomers join together by opening the C=C double bond	condensation reaction with loss of a small molecule (usually water) each time a monomer joins the chain
nature of product	only a single product – the polymer	two products – the polymer plus water (or some other small molecule)
	non-biodegradable	can be biodegradable (nylon takes 40–50 years)
	resistant to acids	PET can be hydrolysed back to monomers by acids or alkalis

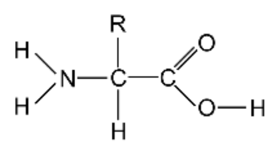
(a) nylon, a polyamide



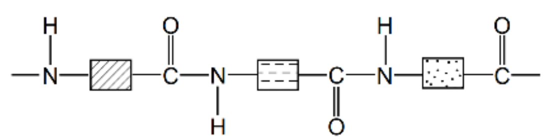
(b) PET, a polyester



- PET can be converted back into monomers and re-polymerised.
- Proteins are natural polyamides and they are formed from amino acid monomers with the general structure (R represents different types of side chain):



- Structure of protein:



- Plastics are non-biodegradable therefore difficult to dispose.
- Problems caused by plastics:
    - disposal in land fill sites
    - accumulation in oceans

    - formation of toxic gases from burning

B12 Experimental techniques and chemical analysis

- Common apparatus:
    - stopwatches
    - thermometers
    - balances
    - burettes
    - volumetric/ graduated pipettes
    - measuring cylinders
    - gas syringes
- Solvent is a substance that dissolves a solute.
- Solute is a substance that is dissolved in a solvent.
- Solution is a mixture of one or more solutes dissolved in a solvent.
- Saturated solution is a solution containing the maximum concentration of a solute dissolved in the solvent at a specified temperature.
- Residue is a substance that remains after evaporation, distillation, filtration or any similar process.
- Filtrate is a liquid or solution that has passed through a filter.
- Simple balance have low resolution, e.g. 0.1g. More complex balance have higher resolution, e.g. 0.001g.
- Temperature can be measured using liquid-in-glass thermometer to the nearest degree. Digital temperature probe have higher resolution, e.g. 0.1°C.
- pH can be measured using either universal indicator paper or more accurately with a digital probe. A digital probe may read to 0.1 or even 0.01 on the pH scale, giving significantly higher resolution.
- **Titration**

1. measure known volume of alkali in a conical flask
2. using pipette
3. methyl orange indicator added in the flask
4. add hydrochloric acid
5. from a burette in small portions, and stir as each portion is added
6. until indicator changes colour
7. record / calculate volume acid added
8. repeat with same volume acid without indicator

- End-point of titration can be identified when the indicator changes colour.
- **Chromatography:**
    - draw a pencil baseline of the chromatography paper
    - add a drop of the dye using a pipette on the baseline.
    - attach the paper to a splint or pencil using a clip and suspend the paper in the solvent
    - remove the paper from the solvent after it has travelled about 80% of the way up the paper.
    - mark the solvent front with a pencil to show how far the solvent moved.
    - dry carefully using a hairdryer or oven.
    - if spots are not visible then spray locating agent.
- <u>A pure substance will give only one spot.</u>

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

- **Crystallisation** → Place the filtrate into an evaporating basin and heat over a water-bath. Allow water to evaporate until crystallisation point or crystals start to form when a glass rod is placed in the solution. pat dry between filter papers.
- **Simple distillation** → Heat the mixture in a distillation flask. The solvent evaporates. The solvent vapour is condensed by passing it down a water-cooled condenser and then collected as the distillate.

- **Fractional distillation** → Heat the mixture in a distillation flask. attach a fractionating column to the flask. The temperature of the fractionating column rises to the boiling point of the most volatile liquid. The other solvents condense back from the beads into the flask until all of the most volatile liquid is collected. The most volatile liquid in the mixture distils first and the least volatile liquid distils last.
- **ALL THE SALTS THAT ARE SOLUBLE IN EXCESS SOLUTION GIVE A SOLUTION OF THAT COLOUR.**

	Lithium, Li^+	Sodium, Na^+	Potassium, K^+	Calcium, Ca^{2+}	Barium, Ba^{2+}	Copper(II), Cu^{2+}
Flame colour	Red	Yellow	Lilac	Orange–red	Light green	Blue–green

Cation	Positive result when NaOH(aq) added
Aluminium, Al^{3+}	White precipitate, this redissolves when an excess of NaOH(aq) is added to give a clear, colourless solution
Ammonium, NH_4^+	No precipitate but produces ammonia on warming
Calcium, Ca^{2+}	White precipitate, this is insoluble in excess NaOH(aq)
Chromium (III), Cr^{3+}	Green precipitate, this will redissolve in excess NaOH(aq)
Copper(II), Cu^{2+}	Light blue precipitate, this is insoluble in excess NaOH(aq)
Iron(II), Fe^{2+}	Green precipitate, this is insoluble in excess NaOH(aq) but starts to turn brown near the surface if left to stand
Iron(III), Fe^{3+}	Red–brown precipitate, this is insoluble in excess NaOH(aq)
Zinc, Zn^{2+}	White precipitate, this will redissolve in excess NaOH(aq) to give a clear, colourless solution

Cation	Positive result when $NH_3(aq)$ added
Aluminium, Al^{3+}	White precipitate; insoluble in excess $NH_3(aq)$
Ammonium, NH_4^+	No reaction
Calcium, Ca^{2+}	No precipitate or very slight white precipitate
Chromium (III), Cr^{3+}	Green precipitate; insoluble in excess $NH_3(aq)$
Copper(II), Cu^{2+}	Light blue precipitate; soluble in excess $NH_3(aq)$ producing a dark blue solution
Iron(II), Fe^{2+}	Green precipitate; insoluble in excess $NH_3(aq)$ but starts to turn brown near the surface if left to stand
Iron(III), Fe^{3+}	Red–brown precipitate; insoluble in excess $NH_3(aq)$
Zinc, Zn^{2+}	White precipitate; redissolve in excess $NH_3(aq)$ to give a clear, colourless solution

Anion	Test	Test result
carbonate (CO_3^{2-})	add dilute hydrochloric acid; test for carbon dioxide gas	effervescence (fizzing), carbon dioxide produced (test with limewater)
chloride (Cl^-) (in solution)	acidify with dilute nitric acid, then add aqueous silver nitrate	white precipitate of silver chloride formed
bromide (Br^-) (in solution)	acidify with dilute nitric acid, then add aqueous silver nitrate	cream precipitate of silver bromide formed
iodide (I^-) (in solution)	acidify with dilute nitric acid, then add aqueous silver nitrate	yellow precipitate of silver iodide formed
sulfate (SO_4^{2-}) (in solution)	acidify with dilute nitric acid (or hydrochloric acid), then add barium nitrate solution	white precipitate of barium sulfate formed
sulfite (SO_3^{2-})	add dilute hydrochloric acid, then add aqueous potassium manganate(VII) solution	decolourises the purple potassium manganate(VII) solution
nitrate (NO_3^-) (in solution)	add aqueous sodium hydroxide, then add aluminium foil and warm carefully; test for ammonia gas	ammonia gas given off (test with moist red litmus)

Gas	Test	Positive result
Hydrogen (H_2)	Lit splint	Produces a squeaky pop
Oxygen (O_2)	Glowing splint	The splint is relit
Carbon dioxide (CO_2)	Bubble through limewater	Turns from clear to cloudy/milky
Chlorine (Cl_2)	Damp litmus paper	Bleached white
Ammonia (NH_3)	Damp red litmus paper	Turns from red to blue
Sulfur dioxide (SO_2)	Solution of acidified potassium manganate(VII)	Decolourised (purple to colourless)

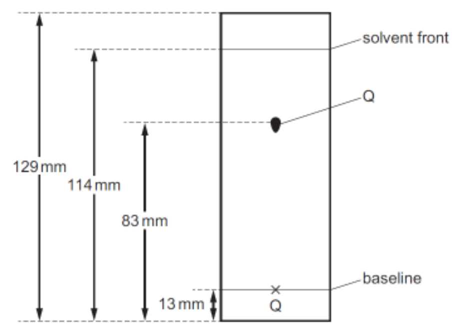
BONUS POINTS

- Ammonium salts react with alkalis to produce **a metal salt, water and ammonia.**
- As the pressure increase MP & BP increase, but for water MP & BP decrease.
- Ammonia react with acid to produce salt.
- Reactivity series mnemonic:
  - Please → Potassium
  - Send → Sodium
  - Cats → Calcium
  - Monkeys → Magnesium
  - And → Aluminium
  - Cute → Carbon
  - Zebras → Zinc
  - In → Iron
  - Their → Tin
  - Lovely → Lead
  - Happy → Hydrogen
  - Cages → Copper
  - Made of → Mercury
  - Silver & → Silver
  - Gold → Gold
- State symbol for water is l (small L).
- Carbon monoxide and Nitrogen oxide (N_2O, NO) is a neutral oxide.
- Nitrates decompose to give metal oxide + nitrogen dioxide + oxygen.
- Group 1 nitrates decompose to give metal nitrite + oxygen.
- Limestone is calcium carbonate. Lime is calcium oxide. Slaked lime is calcium hydroxide.
- Lime is used in desulfurization of flue gases..
- Carboxylic acids don't decolourise bromine water.
- Carbonates decompose to give metal oxide and carbon dioxide.
- Group 2 carbonates decompose easily.
- Hydroxides decompose to give metal oxides and water.
- The higher the Rf value the further the unknown substance travels.
- Ethene is polymerised to give poly(ethene) **NOT** poly(ethane).

- In an equation oxidising agent and reducing agent are present only on the left hand side of the arrow.
- Ionic compounds don't burn in oxygen.
- Mild steel does not have a high resistance to corrosion. It is used in car bodies.
- Water is not produced during cracking.
- The distance moved by the substance is calculated starting from the baseline. The distance moved by the solvent is calculated starting from the baseline. So in this example, subtract 13 from both 83 and 114, and then divide.

3 Substance Q was investigated using chromatography.

The chromatogram is shown. The diagram is not drawn to scale.



What is the R_f value of Q?

A 0.60 **B** 0.64 **C** 0.69 **D** 0.72

- Insoluble salts can only be made by mixing two soluble aqueous salts or acids.
- Changing the concentration or temperature doesn't affect the activation energy.
- Dilute sulfuric acid is not a dehydrating agent. Concentrated sulfuric acid is a dehydrating agent.
- Lead compounds are formed from some types of petrol.
- Oxides of nitrogen are formed from the combustion reactions inside car engines.
- Sulfur dioxide is formed from the combustion of coal.
- Sulfur occurs naturally as element sulfur.
- Electrons **do not** move round the external circuit from the cathode to the anode. **Ions** move round the **internal** circuit from the cathode to the anode.
- More reactive metals are stronger reducing agent.
- Decreasing the concentration of a product favours the side that makes more of that product. In this example, if

u decrease concentration of oxygen equilibrium will shift to side that makes of of that reactant, i.e. equilibrium will shift to the left.

16 The formation of sulfur trioxide from sulfur dioxide is a reversible reaction.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

The forward reaction is exothermic.

Which changes would increase the equilibrium yield of SO_3?

1 increasing the pressure
2 lowering the temperature
3 decreasing the concentration of oxygen

A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 and 3 only

- Down group 1 metals become softer so their melting point and boiling point decrease decrease. Density increase. Reactivity increase.
- During treatment of water for drinking, filtration and chlorination takes place. distillation **doesn't** take place.
- Iodine is a bluish-black solid, yellow/red/brown liquid and violet gas.
- Sulfur mainly occurs in its natural state as sulfur element.
- When we write a charge on ion, compound, element the negative/positive sign comes after the number of charge. Ca^{2+}
- When we write the charge alone without the symbol, we write the negative/positive sign before the number. e.g. +1, -7.
- To convert concentration from mol/dm3 to g/dm3 multiply the concentration by Mr. Example: to convert 0.02 mol of Ca(OH)2 to g/dm3 multiply 0.02 by 74 (Mr of Ca(OH)2) = 1.48.
- Whenever writing reaction for Haber or contact process use **reversible** symbol And also write that the reaction is reversible.
- Silicon(IV) oxide doesn't conduct electricity in molten state.
- Calcium oxide, oxygen and heat is needed to convert iron to steel.
- Group VIII is also called group 0.
- To make a salt by mixing acid and metal oxide, you must heat the acid **First.**
- Carbon dioxide is a waste product in extraction of iron, nitrogen isn't produced during the process.
- Cattel farming produce both carbon dioxide and methane.

- The process of fermentation and addition of stem to ethene to produce ethanol, both use a catalyst. yeast is the catalyst in fermentation.
- Electrons move from anode to cathode in external circuit. Positive ions move from anode to cathode in electrolyte (internal circuit).
- Contact process and Haber process both are exothermic reactions.
- Oxide of sulfur is used as food preservative.
- Stainless steel doesn’t rust.
- Nickle catalyst is used in addition of stem to ethene to produce ethanol.